Kyzyl-Tas - a digital twin of an open-pit copper mine on OpenEgiz

Fictional open-pit Cu-Au mine with a 30 Mt/a concentrator, parametrised from public data for Kazakhstan's Bozshakol and Aktogay operations. Every mine object is an Eclipse Ditto twin; a physics-based engine drives them over MQTT; a 3D interface reads them back from the platform and runs what-if scenarios.



Start (clean Ubuntu with Docker):

```
sudo apt-get update && sudo apt-get install -y make
make up
```

Open

`http://localhost:8000` - 3D mine, no login

`http://localhost:3000` - Grafana: mine dashboard; OpenEgiz Twins/Types (login printed by `make up`)

87 twins · 13 what-if scenarios with Undo · 4 modules · shadow baseline

THE MINE

A real-scale operation, checked against the reference site

- **Pit:** 10 benches (waste cap above, ore below), main and alternate ore ramps, waste ramp, 12 haul-road segments.
- **Fleet:** 14 haul trucks (12× Cat 785C 136 t, 2× Komatsu 830E-5 227 t with AC electric drive), 4 Hitachi EX3600 shovels, 2 drills.
- **Plant:** gyratory crusher 4,186 t/h → 40 kt crushed-ore stockpile → SAG 28 MW + 2 ball mills 22 MW → flotation → tailings.
- **Stockpiles:** ROM pad, crushed ore, high-grade reserve (grade control), waste dump.
- **Site:** 214 MW substation, weather station (Kazakh steppe, down to −38 °C), 2 slope radars, dust stations.

Plant capacities, fleet types, power and climate: KAZ Minerals reports and presentations [S1-S5]; equipment specifications [S11-S14]. Full sourced parameter set: `mine/docs/03_domain.md` .

Twin baseline vs reference (1,000 mine hours, seed 3, `tools/validate.py`)

Quantity	Twin	Reference
Mill feed (with mill outages)	3,245 t/h (28.4 Mt/a)	30 Mt/a design
Copper recovery	79.4 %	79.7 % (2020)
Cu in concentrate per tonne milled	4.10 kg/t	4.04 kg/t (2020)
Truck fleet availability	92.6 %	88-92 % target
Haul cycle	24 min	~22 min (fleet plan)
Unplanned failures / week	24	from all-causes MTBF

BUILT ON OPENEGIZ

Every object is a Ditto twin; all data flows through the platform

mine-engine
Python simulation, 60× real time

→

Mosquitto
Ditto-protocol merge, telemetry/<twin>

→

Eclipse Ditto
87 twins · kz.kyzyltas.mine

→

Telegraf → InfluxDB
history of every feature

→

Grafana
dashboard · Twins/Types pages

mine-ui (nginx + React/three.js) reads the twins from Ditto - REST for the catalogue, server-sent events for live changes - and history from InfluxDB. Its nginx adds the platform credentials server-side, so the browser needs no login.

What-if path: the interface writes `PATCH .../things/kz.kyzyltas.mine:scenario` → Ditto publishes the change on MQTT → the engine applies it → effects return through the equipment twins and the `kpi` twin.

Started by one command. `deploy/compose/mine.yml` adds `mine-setup` (creates the twins), `mine-engine` and `mine-ui`; the Makefile always loads it, so `make up` starts platform and mine together.

Platform fixes made for a clean Ubuntu VM: LF line endings, Telegraf topic filter and no per-command tags (series explosion), mine port in env/preflight/smoke, anonymous read-only Grafana, provisioned mine dashboard.

The first start downloads about 2.5 GB of images (several minutes); later starts take about a minute. `make smoke` checks the whole chain end to end.

MINE MODEL

21 type twins, 66 instance twins, one hierarchy



- **Types:** each instance names its type twin in `definition`; type twins hold instance counts and the feature names and units.
- **Hierarchy** in `_parents`: mine → pit → benches, radars; mine → dispatch → trucks, shovels, drills; mine → plant → crusher, conveyors, mills, flotation, tailings, stockpiles.
- **Material flow** in `feeds`: shovels → ROM pad → crusher → stockpile → SAG → ball mills → flotation → tailings; the substation feeds the plant. Shovels name their `bench`.
- **Live relations:** truck `assigned_shovel` and `road_segment`, road `closed`, grade-control `reclaim_tph`.
- **States:** haul phase, health, queue, throughput, power, grade, drill rate, radar creep, PM10…
- A contract test fails on any missing, extra or wrongly typed feature between engine and twins.

See it: **Twin model** button in the 3D view (no login), or Grafana → OpenEgiz → Twins / Types.

BEHAVIOUR

Physics and process models, linked through the stockpiles

Process	Model
Truck travel	rimpull / grade resistance per segment; ramp, winter and visibility speed caps
Haul cycle	queue → spot → load (passes × swing) → haul → dump (2 bays) → return
Blasting	Kuz-Ram X50/X80 from powder factor
Grinding	SAG throughput from feed size, hardness (Axb), power and ball mills in service; Bond P80; power follows load
Flotation	constant-tails recovery with grind and residence penalties
Stockpiles	mass and grade balance (mass conserved): ROM, crushed, high-grade, waste
Reliability	Weibull wear-out with condition signal + random faults (all-causes MTBF)
Dust	US EPA AP-42 haul-road PM10; no watering below 0 °C
Energy, economics	diesel by class/phase, grid power, CO_2 (0.80 kg/kWh), net smelter revenue

Interlinked. Trucks deliver ore at the grade of the face they loaded → ROM pad → crusher (plus high-grade ore when grade control asks) → 40 kt crushed-ore stockpile → SAG mill → flotation. Pit changes reach the plant only through the stockpiles - so haulage scenarios hit "ore mined" at once and plant output later.

Constraints and back-pressure: one truck under a shovel, two dump bays, winter speed caps, one ball mill down cuts grinding to 60 %, the crusher stops when the crushed pile is full and ore trucks wait when the ROM pad is full, closed ramp → 2.7 km detour, only electric trucks use the trolley line.

Shadow baseline: on every scenario click or Undo, a copy of the live mine (including random streams) continues without the scenario - the deltas measure that scenario only.

WHAT-IF, STEP BY STEP

Shovel S2 failure - the causal chain on the stand



1. `shovel-s2` forced down (hydraulics, 10 mine hours) - red in 3D and on the minimap.
2. Dispatch re-routes S2's trucks to S1 (event log: "rerouted").
3. This screenshot, ~100 mine-minutes after the click: ore mined −30 %, ore grade 0.51 → 0.47 %, contained Cu mined −36 %, haul cycle +31 %. (Slide 7 gives two-hour averages, which include the first minutes.)
4. Grade control reclaims the high-grade stockpile (800 t/h, highlighted).
5. The crushed-ore buffer keeps the mill full: plant copper only −0.5 % so far; over 12 mine hours it falls 2.8 % on average (6.6 % with the fleet-management system offline).

Undo - back to baseline: every `scenario` field returns to its default, S2 returns to service at once, trucks are re-routed back, and the comparison restarts from zero.

WHAT-IF SCENARIOS

13 one-click scenarios, measured against the shadow baseline

Scenario	What changes	First 2 mine hours (average)	Mine hours 2-10
Shovel S2 failure	S2 down 10 mine h	ore −15 %, grade −5 %, cycle +18 %	ore −17 %; Cu output −3 % (buffer, grade control)
Slope radar alarm	main ore ramp closed	ore −16 %, cycle +19 %	ore −30 %, cycle +28 %
Post-blast fume delay	S1, S3 blocked ~150 min	ore −15 %, cycle +19 %	ore −6 %, recovering
Siberian cold snap (−38 °C)	temperature, wind	ore −5 %, cycle +9 %	ore −13 %; road PM10 ≈ ×2
Spring thaw	rolling resistance +2.5 pts	ore −14 %, cycle +27 %	ore −26 %, cycle +39 %
Fleet cut (−2 trucks)	two ore trucks parked	ore −10 %	ore −14 %
Fleet-management system offline	fixed plan, no grade control	≈ 0 %	≈ 0 %; matters during failures (slide 8)
Intensify blast (PF 0.30 → 0.40)	powder factor ×1.33	mill feed, Cu, revenue +11 %	same
Harder ore domain (BWi 14 → 16)	Bond Wi, Axb	mill feed, Cu, revenue −24 %	same; buffer +11 %
Grid power curtailment (mills at 75%)	SAG and ball mills at 75 % power	mill feed, Cu −25 %	grid energy −23 %, CO_2 −21 %
Enable trolley assist	830E trucks on trolley	diesel −4 %	diesel −7 %, ore +5 %
Copper price drop (−25%)	Cu $10,900/t, TC/RC +$80	revenue −19 %	revenue −19 %
Predictive maintenance	maintenance strategy	no change yet	fewer failures over days (slide 8)

Averages against the shadow baseline (seed 3) from `mine/engine/tools/measure_scenarios.py` ; random failures make the live stand vary. Every scenario stays active until ↺ **Undo - back to baseline**. **Custom what-if** sets the scenario twin's values directly (shovel out, trucks parked, crusher, powder factor, dispatch, temperature, ore hardness, mill power, road condition, copper price).

MODULES

Modules that decide while the mine runs - and energy accounting

Dispatch - Haulage and Dispatch

Sends each empty truck to the eligible shovel with the shortest queue; skips broken, closed-off or fume-blocked shovels. Offline, trucks of a broken shovel idle ~30 min until a radio reassignment. With dispatch and grade control, an S2 failure costs **−23 %** ore and **−2.8 %** copper; offline **−28 %** and **−6.6 %** (mean of 8 runs).

Grade control - Geology and Ore Quality

When ROM ore falls below 0.50 % Cu, reclaims the 0.65 % high-grade pile into the crusher feed in the proportion that restores the target (≤ 800 t/h). Seen in `stockpile-hg.reclaim_tph` and "grade control" events.

Predictive maintenance - Equipment Reliability

Health and hours to wear-out are estimated from coolant and oil-iron signals; work orders bring trucks in at the next shift change instead of failing in service. Over 200 mine hours (8 runs): unplanned failures −23 % (191 vs 249); availability about equal, since planned repairs also take trucks out.

Energy and CO_2 - Energy

Accounting, not a control decision: diesel by truck class and phase, grid power for crushing, grinding and trolley lines, cost and CO_2 at 0.80 kg/kWh. Trolley assist (two electric 830E trucks): diesel −4…−7 %, CO_2 about unchanged on the coal-heavy grid (break-even ≈ 0.64 kg/kWh).

HISTORY IN THE PLATFORM

Grafana: current mine vs shadow baseline from InfluxDB



- Provisioned dashboard **"Kyzyl-Tas Mine - KPIs vs baseline"**, Grafana's home page, read-only without login.
- Solid: live mine; dashed grey: shadow baseline.
- Ore mined, haul cycle, mill feed, copper, crushed-ore buffer, grade, revenue, CO_2, diesel, fleet availability, grade-control reclaim, haul-road PM10.
- Every point comes from twin events: Ditto → MQTT → Telegraf → InfluxDB.

DATA AND ASSUMPTIONS

Real sources, synthetic telemetry, stated limits

- **All telemetry is synthetic** - generated by the engine. No company's operational data is used; no third-party dataset is stored.
- **The mine is fictional**; its parameters come from public sources and carry tags: reported [R], vendor [V], literature [L], derived [D], assumed [A].
- Sources: KAZ Minerals reports [S1-S5], equipment specs, Kuz-Ram, Bond, AP-42, Kazakh grid factors and tariffs, 2026 metal prices, reliability studies - listed in `mine/docs/03_domain.md` §12.
- **Assumptions** [A]: road network and pit geometry, failure parameters, high-grade pile, grade-control target, grid factor.

Limitations

- 14 trucks vs Bozshakol's 18: ore demand met, strip ratio ≈ 0.39 vs 0.7 planned.
- Each haul leg timed as one block; per-segment traffic is estimated.
- Price drop changes revenue only; the cut-off grade does not move.
- Dispatch is a shortest-queue rule, not a full LP optimiser.
- Wind, slope-radar velocity and fume delay exist only inside their presets; the other drivers are custom what-if sliders.

Verification: 58 engine tests (physics vs worked examples, mass balance, scenario directions, Undo, baseline copy, mill circuit, dispatch, twin/engine contract); `tools/measure_scenarios.py` and `tools/validate.py` reproduce every number here; `make smoke` checks the running stack.